Assignment - 14

[Section - A]

1. Find modulus and Principal value of Arguement for $-1 + i\sqrt{3}$.

Modulus $|z| = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{4} = 2$

Arguement $(\theta) = \tan^{-1} y/x$

$= \tan^{-1} \frac{\sqrt{3}}{2} = \frac{2\pi}{3}$

2. If $u = 3x - 2xy$, $v = x^2 - y^2 + 3y + c$ then find $f(z)$ in terms

$u = 3x - 2xy$ $\qquad$ $v = x^2 - y^2 + 3y + c$

$u'_x = 3 - 2y$ $\qquad$ $v'_x = 2x$

$f'(z) = u'_x + i v'_x$

$= (3 - 2y) + i(2x)$

put $x = 2$ and $y = 0$

$f'(z) = 3 + i2z$

$f(z) = \int (3 + i2z)\, dz$

$f(z) = 3z + iz^2 + c$

3. Write C-R Equations in Polar Form.

$f(z) = f(r, \theta)$

$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}$ and $\frac{\partial u}{\partial \theta} = -r\frac{\partial v}{\partial r}$

4 Define Complex exponential fn.

$$e^z = e^{x+iy} = e^x(\cos x + i\sin y)$$

5 Discuss nature of the transformation $w = \frac{3z-4}{z-1}$

$$z = \frac{3z-4}{z-1}$$

$$z(z-1) = 3z-4$$

$$z^2 - z = 3z-4$$

$$z^2 - 4z + 4 = 0$$

$$z^2 - 2z - 2z + 4 = 0$$

$$z(z-2) - 2(z-2) = 0$$

$z = 2, 2$ [real and equal] repeated roots

only one repeated point, parabolic transformat

$t = z-2 \qquad z = t+2$

$$w = \frac{3(t+2)-4}{(t+2)-1}$$

$$w = \frac{3t+2}{t+1}$$

$$= 3 - \frac{1}{t+1}$$

$$w = 3 - \frac{1}{z-1}$$

Section - B

Show that fn. $u(x,y) = 4xy - 3x + 2$ is harmonic. Construct corresponding analytic fn. $f(z) = u(x,y) + iv(x,y)$ Express $f(z)$ interms of Complex variable z.

$$u(x,y) = 4xy - 3x + 2$$

$$\frac{\partial u}{\partial x} = 4y - 3 \qquad \frac{\partial u}{\partial y} = 4x$$

$$\frac{\partial^2 u}{\partial x^2} = 0 \qquad \frac{\partial^2 u}{\partial y^2} = 0$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad [\text{laplace Eqn}]$$

$$0 + 0 = 0 \quad [\text{satisfied}]$$

$$dV = \int \left[\frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy\right]$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$dv = -\int 4x\,dx + \int 4y - 3\,dy$$

$$-\frac{4x^2}{2} + \frac{4y^2}{2} - 3y$$

$$= 2y^2 - 2x^2 - 3y$$

$$2(y^2 - x^2) - 3y$$

$$f(z) = u + iv$$

$$f(z) = (4xy - 3x + 2) + i(2y^2 - 3y - 2x^2 + c)$$

$$f'(z) = 4y - 3 + i(-4x)$$

∴ replace x by z and y by 0

$$f'(z) = \int -3 - 4iz \, dz + c$$

$$f(z) = -3z - 2iz^2 + c$$

2. Show that $\cos z = \dfrac{e^{iz} + e^{-iz}}{2}$ and $\sin z = \dfrac{e^{iz} - e^{-iz}}{2i}$

$e^{iz} = \cos z + i\sin z \qquad e^{-iz} = \cos z - i\sin z$

$$e^{iz} + e^{-iz} = 2\cos z$$

$$(\cos z + i\sin z) + (\cos z - i\sin z) = 2\cos z$$

$$2\cos z = 2\cos z$$

Hence Proved

$$e^{-iz} = \cos z - i\sin z$$

$$2\sin z = e^{iz} - e^{-iz}$$

$$= \cancel{\cos z} + i\sin z - \cancel{\cos z} + i\sin z$$

$$2\sin z = 2\sin z$$

3. If $u = 2x^2 + y^2$ and $v = y^2/x$, show that curves $u =$ Constant and $v =$ Constant cut orthogonally at all intersections but that the transformation $w = u + iv$ is not conformal.

$\dfrac{\partial u}{\partial x} = 4x \qquad \dfrac{\partial v}{\partial x} = -y^2/x^2$

$\dfrac{\partial u}{\partial y} = 2y \qquad \dfrac{\partial v}{\partial y} = 2y/x$

For. Orthogonal

$$\boxed{\nabla u \cdot \nabla v = 0}$$

$$4x \cdot \frac{-y^2}{x^2} + 2y \cdot \frac{2y}{x} = 0$$

$$\frac{-(4x \cdot y^2)}{x^2} + \frac{(4y^2)}{x} = 0$$

$$\frac{-4xy^2 + 4xy^2}{x^2} = 0$$

$$0 = 0$$

Hence Proved Orthogonal!

∴ $w = f(z)$ is conformal when it satisfied ~~and~~ C-R Eqns and Jacobian is non-zero.

C-R Eqn $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$$\cancel{4x} = \frac{2y}{x} \quad , \quad 2y = \frac{y^2}{x^2}$$

Only Equal when $y = 2x^2$

only Equal if $y = 0$ or $y = 2x^2$

∴ Therefore, C-R Eqns do not hold everywhere
Thus $f(z)$ is not conformal.

Part - C

1 show that the fn. $f(z) = u + iv$ where

$$f(z) = \frac{x^3(1+i) - y(1-i)}{x^2+y^2}, \quad z \neq 0$$

$$= 0 \qquad z = 0$$

Satisfies the CR eqn @ $z = 0$ Is the fn. analytic @ $z = 0$? Justify your answer.

$$f(z) = \frac{x^3(1+i) - y^3(1+i)}{x^2+y^2}, \quad z \neq 0, \quad f(0) = 0$$

$$f(z) = u + iv$$

$$u = \frac{x^3 - y^3}{x^2+y^2} \qquad v = \frac{x^3 + y^3}{x^2+y^2}$$

Checking CR Eqns @ $z = 0$

at origin

$u_x(0,0) = 0$, $u_y(0,0) = 0$, $v_x(0,0) = 0$

$v_y(0,0) = 0$

CR Eqns are satisfied @ $z = 0$

Checking analiticity

$$f'(0) = \lim_{z \to 0} \frac{f(z) - f(0)}{z} = \lim_{z \to 0} \frac{f(z)}{z}$$

$y = 0$

$$f(z) = \frac{x^3(1+i)}{x^2} = x(1+i) \Rightarrow \frac{f(z)}{z} = 1+i$$

$x = 0$ $$f(z) = \frac{-y^3(1-i)}{y^2} = -y(1-i) \Rightarrow \frac{f(z)}{z} = -(1-i)i$$

$$= \underline{1+i}$$

Diff value @ every point. Hence,

Fn is not analytic @ $z = 0$.

2. If $f(z) = u + iv$ is an analytic fn of $z = x + iy$ and $u + v = \dfrac{2\sin 2x}{e^{2y} + e^{-2y} - 2\cos 2x}$

Find $f(z)$ in terms of z.

$$u + v = \frac{2\sin 2x}{e^{2y} + e^{-2y} - 2\cos 2x}$$

$$e^{2y} + e^{-2y} - 2\cos 2x = 2(\cosh 2y - \cos 2x)$$

So, $$u + v = \frac{\cancel{2}\sin 2x}{\cancel{2}(\cos 2y - \cos 2x)} = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

using identity

$$\cot z = \frac{\sin 2x + i\sinh 2y}{\cosh 2y - \cos 2x}$$

Real + Imag part $\Rightarrow u + v = Re(\cot z) + Im(\cot z)$

Thus $$f(z) = \frac{1}{2}\cancel{(1+i)}\cot z + \frac{c}{1+i}$$

3. Find the billinear transformation which maps the points $z = 1, i, -1$ into the points $z = i, 0, -i$. Hence find image of $|z| < 1$,

Soln $\rightarrow$ $z = 1, i, -1 \longrightarrow w = i, 0, -i$

For billinear transformation

$$\frac{(w - w_1)\cancel{(w_2 - w_3)}}{(w - w_3)(w_2 - w_1)} = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}$$

$$\frac{(w-i)\,i}{(w+i)(-i)} = \frac{(z-1)(i+1)}{(z+1)(i-1)}$$

$$\frac{-(w-i)}{w+i} = -\frac{(z-1)(1+i)}{(z+1)(i-1)}$$

$$\frac{i+1}{i-1} = -i$$

$$\frac{w-i}{w+i} = i\cdot\frac{z-1}{z+1}$$

$$(w-i)(z+i) = (w+i)(i)(z-1)$$

$$(w-i)(z+1) = wz + w - iz - i$$

$$i(wz - w + iz - i) = iwz - iw + i^2 z - i^2$$

$$= i\cdot iwz - iw - z + 1$$

$$w = [(1-i)z + (1+i)] = (i+1) - (1-i)z$$

$$w = \frac{(1+i) - (1-i)z}{(1-i)z + (1+i)}$$

$$\boxed{w = \frac{i-z}{z+i}}$$

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